

## MCKINNEY, TX, USA

WMO: 722541

Lat: 33.190N

Lon: 96.591W

Elev: 179

StdP: 99.20

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 53914

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -6.2       | -3.8 | -12.7                         | 1.3 | -2.4 | -10.2 | 1.6 | 0.9  | 11.9                  | 11.2 | 11.0 | 12.1 | 3.2                   | 0    | 0.472 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 12.1                   | 38.0            | 23.7 | 37.1 | 23.9 | 35.8 | 23.9 | 26.0                | 32.8 | 25.5 | 32.5 | 25.1 | 32.0 | 4.0                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.3                            | 19.6 | 28.2 | 23.8 | 19.1 | 27.9 | 23.2 | 18.4 | 27.6 | 81.4          | 33.1 | 79.1 | 32.8 | 77.3 | 32.1 | 29.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| (a)               | (b)  | (c) | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| 10.7              | 9.2  | 8.3 | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
|                   |      |     | -9.9                       | 40.0 | 2.1                | 2.0 | -11.5                                              | 41.4 | -12.7      | 42.5 | -13.9      | 43.6 | -15.5      | 45.0 |  |  |
|                   |      |     | -10.7                      | 27.0 | 1.9                | 1.0 | -12.1                                              | 27.7 | -13.2      | 28.3 | -14.2      | 28.8 | -15.6      | 29.5 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 18.5   | 7.1   | 9.3   | 13.7  | 18.0  | 22.5  | 27.0  | 29.0  | 29.2  | 25.2  | 19.0  | 13.1  | 7.9  |
|                                                                               | DBStd       | 8.97   | 5.42  | 5.87  | 5.12  | 4.22  | 3.58  | 2.39  | 2.20  | 2.50  | 3.51  | 4.64  | 5.21  | 5.35 |
|                                                                               | HDD10.0     | 360    | 123   | 78    | 22    | 2     | 0     | 0     | 0     | 0     | 0     | 2     | 28    | 106  |
|                                                                               | HDD18.3     | 1373   | 351   | 258   | 157   | 56    | 9     | 0     | 0     | 0     | 2     | 48    | 168   | 324  |
|                                                                               | CDD10.0     | 3450   | 32    | 57    | 138   | 242   | 388   | 510   | 590   | 594   | 455   | 282   | 122   | 41   |
|                                                                               | CDD18.3     | 1422   | 1     | 3     | 14    | 46    | 139   | 260   | 332   | 336   | 207   | 69    | 12    | 2    |
|                                                                               | CDH23.3     | 16319  | 6     | 20    | 92    | 341   | 1252  | 2926  | 4287  | 4393  | 2262  | 654   | 78    | 10   |
|                                                                               | CDH26.7     | 8098   | 0     | 2     | 10    | 60    | 423   | 1410  | 2383  | 2506  | 1087  | 207   | 11    | 1    |
| Wind                                                                          | WSAvg       | 3.6    | 3.8   | 4.1   | 4.3   | 4.5   | 3.9   | 3.6   | 3.1   | 2.8   | 2.8   | 3.2   | 3.7   | 3.7  |
| Precipitation                                                                 | PrecAvg     | 1041   | 71    | 75    | 93    | 95    | 123   | 92    | 57    | 60    | 82    | 117   | 83    | 84   |
|                                                                               | PrecMax     | 1373   | 154   | 222   | 224   | 215   | 205   | 283   | 169   | 149   | 201   | 261   | 208   | 232  |
|                                                                               | PrecMin     | 579    | 1     | 12    | 0     | 29    | 46    | 12    | 0     | 1     | 13    | 11    | 13    | 9    |
|                                                                               | PrecStd     | 221    | 44    | 55    | 54    | 51    | 54    | 70    | 44    | 40    | 48    | 77    | 58    | 56   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 24.0  | 26.0  | 28.0  | 30.8  | 34.1  | 37.1  | 39.4  | 40.5  | 38.3  | 32.9  | 28.0  | 24.8 |
|                                                                               |             | MCWB   | 15.6  | 16.9  | 18.4  | 21.5  | 23.6  | 24.3  | 23.2  | 23.1  | 22.7  | 21.6  | 19.1  | 17.9 |
|                                                                               | 2%          | DB     | 21.7  | 23.1  | 25.9  | 28.1  | 32.3  | 35.5  | 37.8  | 38.6  | 35.8  | 31.1  | 25.8  | 21.9 |
|                                                                               |             | MCWB   | 15.4  | 16.2  | 18.1  | 20.3  | 23.0  | 23.9  | 24.0  | 23.5  | 22.8  | 21.6  | 18.4  | 16.4 |
|                                                                               | 5%          | DB     | 19.1  | 21.3  | 23.9  | 26.9  | 30.8  | 34.1  | 36.6  | 37.4  | 33.8  | 29.0  | 23.6  | 19.6 |
|                                                                               |             | MCWB   | 14.1  | 16.0  | 17.3  | 19.8  | 22.6  | 23.9  | 24.1  | 23.9  | 22.7  | 20.4  | 17.7  | 15.4 |
|                                                                               | 10%         | DB     | 17.2  | 19.1  | 22.2  | 25.4  | 29.0  | 32.9  | 35.2  | 36.1  | 32.4  | 27.2  | 21.7  | 17.4 |
|                                                                               |             | MCWB   | 12.6  | 14.7  | 16.5  | 18.9  | 22.0  | 23.8  | 24.2  | 23.9  | 22.6  | 19.8  | 16.9  | 13.4 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 18.9  | 20.0  | 20.8  | 23.5  | 25.3  | 26.7  | 26.8  | 26.5  | 25.6  | 24.4  | 21.7  | 19.7 |
|                                                                               |             | MCDB   | 21.6  | 22.9  | 25.2  | 27.7  | 31.2  | 33.0  | 33.4  | 33.2  | 31.7  | 28.7  | 25.0  | 22.2 |
|                                                                               | 2%          | WB     | 17.5  | 18.8  | 19.7  | 22.1  | 24.5  | 25.8  | 25.9  | 25.8  | 24.9  | 23.4  | 20.2  | 18.2 |
|                                                                               |             | MCDB   | 19.4  | 21.5  | 23.6  | 26.3  | 29.7  | 32.1  | 32.9  | 33.1  | 30.9  | 27.9  | 23.3  | 20.7 |
|                                                                               | 5%          | WB     | 15.4  | 17.4  | 18.8  | 21.3  | 23.8  | 25.2  | 25.5  | 25.3  | 24.3  | 22.6  | 19.1  | 16.5 |
|                                                                               |             | MCDB   | 18.3  | 19.8  | 22.6  | 25.1  | 28.7  | 31.5  | 32.6  | 32.9  | 30.1  | 26.8  | 22.4  | 18.6 |
|                                                                               | 10%         | WB     | 12.7  | 15.5  | 17.7  | 20.4  | 23.1  | 24.7  | 25.1  | 24.9  | 23.8  | 21.6  | 17.7  | 14.1 |
|                                                                               |             | MCDB   | 16.0  | 18.6  | 21.2  | 23.9  | 27.7  | 30.7  | 32.1  | 32.3  | 29.4  | 25.4  | 20.8  | 16.8 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 12.0  | 11.6  | 11.6  | 11.5  | 10.9  | 10.7  | 11.5  | 12.1  | 12.2  | 12.5  | 11.9  | 11.4 |
|                                                                               |             | MCDBR  | 15.8  | 14.0  | 12.9  | 12.4  | 12.2  | 12.0  | 13.1  | 14.1  | 13.6  | 13.9  | 13.4  | 14.2 |
|                                                                               |             | MCWBR  | 10.1  | 9.0   | 7.3   | 6.6   | 5.0   | 3.7   | 3.4   | 3.7   | 4.2   | 6.0   | 7.5   | 9.4  |
|                                                                               | 5% WB       | MCDBR  | 12.8  | 11.8  | 10.8  | 9.9   | 10.2  | 10.3  | 11.1  | 11.5  | 10.6  | 10.5  | 10.7  | 11.4 |
| MCWBR                                                                         |             | 9.9    | 8.9   | 7.2   | 6.4   | 5.2   | 3.9   | 3.7   | 3.5   | 3.8   | 5.3   | 7.7   | 9.3   |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.314  | 0.327 | 0.358 | 0.394 | 0.428 | 0.440 | 0.443 | 0.446 | 0.400 | 0.357 | 0.337 | 0.318 |      |
|                                                                               | taud        | 2.494  | 2.469 | 2.402 | 2.322 | 2.289 | 2.287 | 2.316 | 2.294 | 2.370 | 2.475 | 2.477 | 2.488 |      |
|                                                                               | Ebn at Noon | 904    | 926   | 919   | 895   | 863   | 848   | 843   | 837   | 864   | 881   | 869   | 877   |      |
|                                                                               | Edh at Noon | 86     | 98    | 113   | 128   | 133   | 133   | 129   | 130   | 115   | 96    | 87    | 82    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 2.80   | 3.28  | 4.36  | 5.31  | 5.74  | 6.49  | 6.62  | 6.09  | 5.11  | 4.00  | 2.99  | 2.46  |      |
|                                                                               | RadStd      | 0.28   | 0.50  | 0.34  | 0.40  | 0.53  | 0.65  | 0.43  | 0.40  | 0.47  | 0.59  | 0.38  | 0.31  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)     | (m)     | (n)     |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (47 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page